No Need to Be So Big

Milestone 2: datasets and exploratory analysis

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1 Datasets

ShallKnow [1] is evaluated on three fact-checking benchmarks: BPDP [2], FactBench Mix [3] and FAVEL-DS [4]. Each of them is a set of reified RDF statements (subject, predicate, object) with a truth label, split into train and test. The authors published all three on Zenodo already mapped from DBpedia to Wikidata, and this is the version we use, since our goal is to rerun their pipeline with smaller language models and compare the results with theirs. The analysis is in `notebooks/01_eda.ipynb` in our repository.

Before looking at the content we compared the files with Table 1 of the paper (Table 1). BPDP and FAVEL-DS match exactly. FactBench does not: the train file has 680 statement IRIs instead of 1119, and 345 of them have two or more subjects, objects and truth values attached. In the original FactBench the true and the false statements are stored in two folders with the same file names (e.g. `birth_00005.ttl` exists in both), the file names became the statement IRIs and the two versions were merged into one. It is not possible to tell afterwards which subject belongs to which object. Only 335 statements per split are intact and almost all of them are true (Figure 1).

We therefore use FAVEL-DS as our main dataset and BPDP for quick tests, and leave FactBench out. Both remaining datasets come from the paper, so our numbers stay comparable with the original results, and they are quite different from each other, which we describe below.

	paper (true/false)		Zenodo file, intact statements	
dataset	train	test	train	test
BPDP	100/100	103/103	100/100	103/103
FactBench Mix	633/486	637/492	309/26	306/29
FAVEL-DS	380/385	163/164	380/385	163/164

Table 1: Dataset sizes reported in the paper and what we found in the published files.

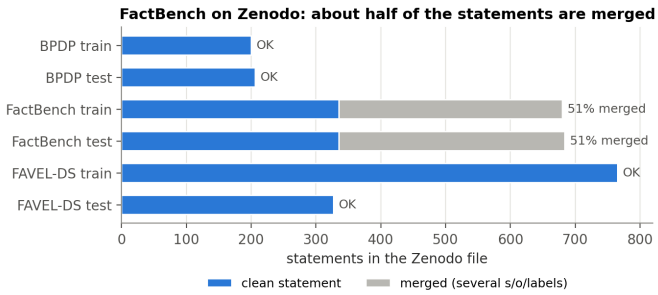


Figure 1: Intact and merged statements per file.

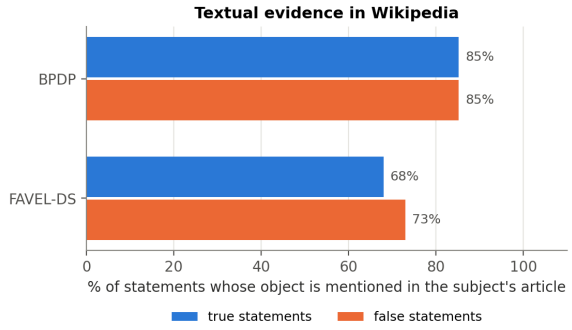


Figure 2: Share of statements whose object is mentioned in the subject's Wikipedia article.

2 Relations, entities and negative examples

BPDP has only two relations, place of birth and place of death, with 406 statements about 245 entities. FAVEL-DS has 11 relations (people, organisations and creative works, e.g. award received, headquarters, director, cast member), 1092 statements and 1181 entities (Figure 3). Both are balanced, every relation has about as many false statements as true ones.

The way the false statements were made differs a lot. For each false statement we looked for a true statement in the same dataset that differs in only one position. In BPDP all 203 false statements are true ones with the predicate swapped: the place of death is given as the place of birth and the other way round (A. E. van Vogt *born in* Hollywood). Both entities really are related, only the relation is wrong. In FAVEL-DS 519 of 549 false statements replace the object with another entity of the right type, which is the more usual setup. Test and train share 54% of the entities in FAVEL-DS and only 12% in BPDP.

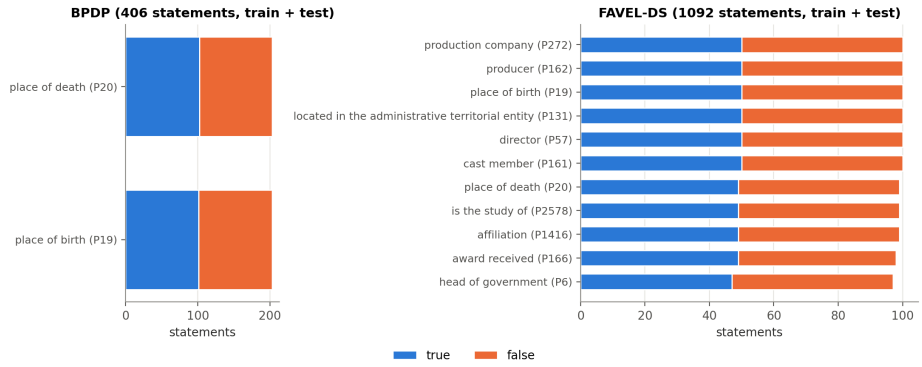


Figure 3: Relations in both datasets, train and test together.

3 Wikipedia as the text source

ShallKnow takes its text from English Wikipedia, starting from the articles of the subject and the object. 97.7% of the FAVEL-DS entities and 92.2% of the BPDP entities have an English article. The missing ones in BPDP are not obscure places though. London, Rome, Cambridge, Sydney, Copenhagen and Cairo were mapped to wrong Wikidata items, e.g. `Q103177460` is "London (Samian Ware Discovery Site)", an archaeological record, and 26 BPDP statements use it. This looks like a side effect of the fuzzy DBpedia-to-Wikidata mapping done by the authors. It affects 10% of entity mentions in BPDP and 2% in FAVEL-DS, and for these statements a path-based fact checker searches for paths to the wrong node.

We also checked whether the text separates true from false statements at all. For all BPDP statements and a random sample of 200 FAVEL-DS statements we tested whether the subject's article mentions the object (Figure 2). In BPDP the answer is the same for true and false statements (85% vs 85%), which follows from how the negatives were made. In FAVEL-DS the object of a false statement is mentioned even a bit more often (73% vs 68%), because the replacement objects are related entities of the same type. Simple co-occurrence is not enough, the relation has to be extracted correctly, and that is exactly the step done by the LLM that we want to make smaller.

4 Cost of the pipeline

The text simplification step calls the LLM once for every article linked from the subject and object articles. On a sample of 60 articles per dataset the median number of links is 508 for BPDP and 247 for FAVEL-DS, with some articles over 2000. In our test runs `deepseek-r1:14b` needed about 7 s per call. Depending on how many articles are shared between statements, a full run with one model would take between 47 and 564 hours for BPDP and between 162 and 826 hours for FAVEL-DS. Judging by the numbers in the paper, the authors processed around 350k articles in total.

5 Conclusions for the next steps

FactBench is not usable in its published form, so we continue with FAVEL-DS and BPDP. Since we want to compare about six models, running the full pipeline for each is out of reach. We will fix one subset of true statements, cap the number of linked articles per entity and run every model on exactly the same input. BPDP is hard by design (only the relation is wrong) and has some questionable labels and wrong entity ids, so we expect small differences there, which is in line with the paper (at most +0.04 AUROC on BPDP). FAVEL-DS, where the paper reports the largest gains, should show the differences between models more clearly.

References

[1] U. Qudus, N. Pokharel, M. Röder, A.-C. Ngonga Ngomo. No Need to Be a Know-It-All: Fact Checking with Shallow Knowledge. ESWC 2026, pp. 436–456.

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[4] U. Qudus, F. L. T. Pekarou, A. A. M. da Silva, M. Röder, A.-C. Ngonga Ngomo. FAVEL: Fact Validation Ensemble Learning. EKAW 2024, pp. 209–225.